There are **two ways** to get information:

- You collect it yourself → **Primary Data**
- Someone else already collected it → **Secondary Data**

---

1. Primary Data

Definition

Primary data is **original data collected directly by you for a specific purpose.**

Think of it as **freshly cooked food**.

You made it yourself.

Example

Your college wants to know:

"How many students use AI tools?"

You create a Google Form and ask 500 students.

The answers you receive are **Primary Data**.

Real-Life Examples

- Conducting surveys
- Taking interviews
- Observations
- Experiments
- Measuring temperature yourself
- Counting vehicles on a road
- Recording attendance

College Example

A professor wants to know:

Which programming language do FYBCA students prefer?

He creates a questionnaire.

Students answer:

- Python
- Java
- C++
- PHP

These responses are Primary Data.

Advantages

✅ Accurate for your problem

✅ Latest information

✅ More reliable (if collected properly)

✅ Full control over collection

Disadvantages

❌ Takes time

❌ Expensive

❌ Requires planning

❌ Requires manpower

2. Secondary Data

Definition

Secondary data is **data already collected by someone else and reused by you.**

Think of it as **ordering food from a restaurant.**

Someone else already prepared it.

Example

Instead of conducting your own survey, you download a government report.

That report is Secondary Data.

Sources

- Government reports
- Books
- Newspapers
- Research papers
- Census reports
- Company reports
- Kaggle datasets
- World Bank
- WHO
- IMF
- University research

College Example

You are making a project on India's population.

Instead of surveying 1.4 billion people,

you download Census data.

That is Secondary Data.

Advantages

✅ Cheap

✅ Quick

✅ Large datasets available

✅ Easy to access

Disadvantages

❌ May be old

❌ May not match your problem exactly

❌ Unknown data quality

❌ Limited control

Primary vs Secondary Data

Feature	Primary Data	Secondary Data
Collected By	You	Someone else
Cost	High	Low
Time	More	Less
Accuracy	High for your purpose	Depends on source
Control	Full	None
Example	Survey	Census report

Memory Trick

Primary = Produce Yourself

Secondary = Someone Already Did It

Example

Suppose you want to know:

"How many students use ChatGPT?"

Primary

Ask students directly.

Secondary

Read an existing report.

---

3. Open Data Sources

What is Open Data?

Open Data is **data that anyone can access, use, and share**, usually for free, subject to its license.

Imagine a huge public library where everyone can borrow books.

Open data works similarly for datasets.

Why Open Data?

Governments and organizations collect massive amounts of information.

Instead of keeping it private, they publish it so that:

- Students can learn
- Researchers can analyze
- Developers can build apps
- Businesses can innovate
- Citizens can understand public services

Examples of Open Data

- Population statistics
- Rainfall records
- Air pollution
- COVID-19 data
- Road accidents
- Railway schedules
- Weather
- Crop production
- Education statistics
- Election results

Popular Open Data Portals

1. Open Government Data Platform India

Contains:

- Government datasets
- Agriculture
- Education
- Health
- Transport
- Census
- Economy

Ideal for Indian student projects.

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2. Kaggle

Contains

- Machine Learning datasets
- AI datasets
- Finance
- Sports
- Movies
- Images
- Text datasets

Very popular among data scientists.

3. World Bank Open Data

Contains

- GDP
- Poverty
- Population
- Economy
- Health
- Education

---

4. Data.gov

Provides thousands of public datasets from U.S. government agencies.

Examples of Student Projects Using Open Data

- Weather prediction
- Pollution dashboard
- Crime analysis
- Hospital management statistics
- Stock market visualization
- COVID dashboard
- Tourism analysis
- Smart city analytics

4. APIs (Application Programming Interfaces)

What is an API?

Imagine a restaurant.

- You = Customer
- Kitchen = Database
- Waiter = API

You don't go into the kitchen.

You tell the waiter what you want.

The waiter brings your food.

The API works exactly like that:

```
Application
      |
      ▼
     API
      |
      ▼
Database / Server
```

Example

A weather app asks:

```
What is today's temperature?
```

The API replies:

```
32°C
Humidity: 70%
```

Real Examples

- Weather apps
- Google Maps
- Cricket score apps
- Stock market apps
- Payment apps
- Food delivery apps

API Response Example (JSON)

```
{
  "city": "Surat",
  "temperature": 32,
  "humidity": 70
}
```

Computers can easily read this structured data.

Advantages of APIs

- Real-time data
- Automatic updates
- Easy integration
- Fast and reliable
- Reduces manual work

---

5. Web Scraping Basics

What is Web Scraping?

Web Scraping is the process of **automatically extracting information from web pages** using software instead of copying it manually.

Imagine you need the prices of 5,000 laptops.

Manual Method

Open every page.

Copy each price.

It could take several days.

Web Scraping

A program visits every page, collects the prices, and saves them into a spreadsheet in minutes.

Example

Website

```
Laptop A - ₹45,000
```

```
Laptop B - ₹52,000
```

```
Laptop C - ₹61,000
```

After scraping

Laptop	Price
A	45000
B	52000
C	61000

Basic Steps in Web Scraping

1. Send a request to a website.
2. Download the webpage (HTML).
3. Parse the HTML to locate the required information.
4. Extract the desired data (e.g., product names or prices).
5. Save it to a file such as CSV, Excel, or a database.

Applications of Web Scraping

- Price comparison websites
- Job portals
- News aggregation
- Market research
- Sports statistics
- Real estate analysis
- E-commerce analytics